



Copper Dynamics in the Aquatic Environment Under Ocean Acidification and Contamination by Microplastics

Luísa Silva Poersch[✉] · Sanye Soroldoni[✉] · Fernanda Chaves Lopes[✉] ·
Camila De Martinez Gaspar Martins[✉] · Marco Tadeu Grassi[✉] · Brenno Januário da Silva ·
Maurício Garcia de Camargo[✉] · Rodrigo Kerr[✉] · Grasiela Lopes Leães Pinho[✉]

Received: 11 November 2025 / Accepted: 1 April 2026
© The Author(s) 2026

Abstract Microplastics are pollutants that cause great concern due to their known effects on biota and growing accumulation in the oceans. Microplastics can act as vectors of copper (Cu) ions, potentially altering their dynamics and bioavailability in the marine environment through adsorption processes, which are influenced by several factors, including pH. Given that ocean acidification is a current process that can be further aggravated through many processes, including an increase in anthropogenic CO₂ emissions, one of the aims of this study was to evaluate the behaviour of Cu (II) in the presence of photodegraded microplastics (polyethylene terephthalate (PET) in two different reduced pH scenarios

according to the Shared Socioeconomic Pathway predicted for 2100 by IPCC. The other aim was to verify chemical changes in the carbonate system and Cu (II) speciation in both reduced pH scenarios predicted for 2100 by IPCC, focusing on inorganic complexation. At the moderate acidification (pH 7.88) treatment, the concentration of dissolved Cu (II) decreased after a 48-h exposition to PET microplastics compared to the initial concentration, primarily due to inorganic complexation and adsorption onto natural particles rather than adsorption onto microplastics, which exhibited a limited capacity to adsorb Cu²⁺ under the experimental conditions. Conversely, in the intense acidification treatment (pH 7.59), the concentration of dissolved Cu (II) increased, likely due to the release of previously complexed Cu (II) and the reduced adsorption capacity of the microplastics, as Cu²⁺ competes with the increased H⁺ concentrations for the sorption

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11270-026-09440-1>.

L. Silva Poersch (✉) · S. Soroldoni · R. Kerr ·
G. Lopes Leães Pinho
Programa de Pós-graduação em Oceanologia, Instituto
de Oceanografia, Universidade Federal do Rio Grande
(FURG), Av. Itália Km 8, Rio Grande, Rio Grande do Sul,
474, 96203-900, Brazil
e-mail: luisa.poersch@gmail.com

S. Soroldoni
e-mail: sanyesoroldoni@gmail.com

R. Kerr
e-mail: rodrigokerr@furg.br

G. Lopes Leães Pinho
e-mail: grasielapinho@furg.br

L. Silva Poersch · S. Soroldoni · G. Lopes Leães Pinho
Laboratório de Microcontaminantes Orgânicos e
Ecotoxicologia Aquática, Instituto de Oceanografia,
Universidade Federal do Rio Grande (FURG), Av.
Itália, Km 8, Rio Grande, Rio Grande do Sul, 474,
CEP: 96203-900, Brazil
e-mail: sanyesoroldoni@gmail.com; e-mail: grasielapinho@furg.br

sites on the surfaces of microplastics. These results indicate that under a more intense acidification scenario, the capacity of microplastics to act as vectors of Cu (II) could be insufficient, potentially increasing the concentration of free ionic-Cu and increasing its availability and consequent toxicity to marine biota.

Keywords Adsorption · Climate change · Metal · PH · Plastic pollution

1 Introduction

Microplastics are synthetic polymers particles ranging from 1 μm to 5 mm (Hanvey et al., 2017). Since their first report in the marine environment (Carpenter and Smith, 1972), their abundance in this environment has intensified due to the high expansion of global plastic production and inadequate waste management (Barnes et al., 2009). Owing to their durability and resistance to degradation, microplastics persist in aquatic environments and accumulate in marine and coastal environments (Lusher et al., 2017).

According to Guo and Wang (2019), the most frequently found microplastics in the ocean include polyethylene (PE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), polyamide (PA) and polyethylene terephthalate (PET). In 2023, global plastic production reached 413.8 Mt, of which PET represents a significant fraction of global plastic production (*i.e.* (6.2% of this total) (Plastics Europe, 2024) and is widely used in the packaging applications (Geyer et al., 2017). Field studies have reported PET among the dominant polymer types in the marine environment (Enamul Kabir et al., 2020), which demonstrates their high environmental persistence. PET

plastics are classified as thermoplastic, semicrystalline and hydrophobic polymers (Andrady, 2017; Guo and Wang, 2019). They also have a density superior to marine water's and therefore tend to sink in the water column (Andrady, 2017; Hale et al., 2020).

Although they are considered contaminants in themselves, microplastics can interact with other contaminants present in water due to their physicochemical characteristics and large surface area (Liu et al., 2021), such as persistent organic pollutants as well as metallic species (Harmon, 2018). One example is copper (Cu), an essential trace metal in marine biota at low concentrations due to its role in certain enzyme systems or as an essential element of hemocyanin in invertebrates (Pilson, 2013). Nonetheless, when the ambient concentration surpasses tolerance thresholds, Cu can have toxic effects on aquatic organisms (Millero et al., 2009). From an ecotoxicological perspective, copper is known to exert sublethal effects at low $\mu\text{g L}^{-1}$ levels, particularly depending on its speciation. Toxicity is primarily associated with the free Cu^{2+} ion rather than total dissolved copper. Therefore, the expansion of anthropogenic sources of copper, such as mining activities, metal and electrical manufacturing, agriculture and specially antifouling paints in the marine environment (USEPA, 2026; Soroldoni et al., 2018) can increase the metal's toxicity and its potential to interact with plastic particles, such as PET-based microplastics.

Microplastics can interact with Cu through adsorption, a chemical or physical process in which atoms, molecules or ions adhere to solid surfaces (Lusher et al., 2017). The adsorption mechanisms between microplastics and metals can be characterized by electrostatic interactions, van der Waals forces, and π – π interactions, among others (Liu et al., 2021).

F. Chaves Lopes · C. De Martinez Gaspar Martins
Programa de Pós-Graduação em Ciências Fisiológicas,
Instituto de Ciências Biológicas, Universidade Federal
do Rio Grande (FURG), Av. Itália Km 8, Rio Grande,
Rio Grande do Sul, 474, 96203-900, Brazil
e-mail: fernandach.lopez@gmail.com

C. De Martinez Gaspar Martins
e-mail: camilaos@hotmail.com

M. T. Grassi
Departamento de Química, Universidade Federal
do Paraná (UFPR), XV de Novembro, 1299,
Curitiba, Paraná, 19032, CEP: 81531-980, Brazil
e-mail: mtgrassi@ufpr.br

B. Januário da Silva · R. Kerr
Laboratório de Estudos dos Oceanos e Clima, Instituto
de Oceanografia, Universidade Federal do Rio Grande
(FURG), Av. Itália, Km 8, Rio Grande, Rio Grande do Sul,
474, 96203-900, Brazil
e-mail: brennojanuario@gmail.com

M. Garcia de Camargo
Laboratório de Ecologia de Invertebrados Bentônicos,
Instituto de Oceanografia, Universidade Federal
do Rio Grande (FURG), Av. Itália, Km 8, Rio Grande,
Rio Grande do Sul, 474, 96203-900, Brazil
e-mail: camargofurg@gmail.com

Microplastics can have negative charges and thus attract metallic cations if their point of zero charge (pH_{zcp}) is lower than the pH of the environment in which they are present (Li et al., 2019).

Metal adsorption onto microplastics can be influenced by many factors, such as the contact time between the two contaminants, ionic forces, plastic particle sizes, microplastic dosages, ageing levels of microplastics, and environmental parameters such as water temperature and pH (Cao et al., 2021; Liu et al., 2021). Photodegradation is one of the main ageing processes of microplastics and occurs through the exposure of microplastics to ultraviolet (UV) radiation from the sun. The artificial photodegradation of PET microplastics reportedly enhances their capacity to adsorb Cu and zinc (Zn) compared with that of pristine microplastics (Wang et al., 2020). According to the authors, UV radiation increased both the surface areas of the particles through the formation of irregularities and oxygen-containing groups, which contributed to enhancing the affinity between the metals and aged microplastics, facilitating the adsorption process.

Among environmental factors, pH is particularly relevant because it simultaneously controls microplastics surface charge and metal speciation in solution (Gao et al., 2024) acidic conditions may promote protonation of surface functional groups and increase competition between H^+ and metal cations, whereas higher pH promotes deprotonation and enhances electrostatic interactions and surface complexation (Tang et al., 2025; Wang et al., 2024). Relatively few studies have evaluated the adsorption of metal ions onto PET microplastics. Wang et al. (2022) have observed increased Cu (II) adsorption with rising pH from 5 to 7, followed by a decline when the pH surpassed 7, reaching its lowest value at pH 8. On the other hand, Sekar and Sundaram (2026) observed that adsorption of Pb (II) onto PET microplastics was strongly altered by environmental factors, with higher pH (9) and moderate temperature (15–25 °C) favouring the metal uptake on MPs. In the case of PET microplastics, the combined influence of pH variation and changes in copper speciation under altered carbonate chemistry remains insufficiently investigated, particularly in the context of ocean acidification.

As the ocean's pH is suffering a decrease due to acidification, it is expected that this process may affect the metal adsorption capacity of microplastics

in the marine environment. Ocean acidification is driven by the diffusion of elevated atmospheric anthropogenic carbon dioxide (CO_2) into surface waters, altering the dynamics of the marine carbonate system (Doney et al., 2009). The absorbed CO_2 quickly reacts with water to form carbonic acid (H_2CO_3), which releases hydrogen protons (H^+) and consequently increases the concentration of bicarbonate ions (HCO_3^-) and the total dissolved inorganic carbon (DIC) level, whereas there is a decrease in the concentration of carbonate ions (CO_3^{2-}) and the water pH (Gattuso & Hansson, 2011). Therefore, owing to the increase in CO_2 emissions over time, the average pH of the ocean has decreased from 8.2 to approximately 8.1 since the industrial period. According to IPCC, the ocean's pH can decrease even further depending on the Shared Socioeconomic Pathways taken by worldwide society: in the intermediate scenario (SSP2-4.5), if CO_2 emissions begin to decline by 2050, the ocean's mean pH can decrease 0.17 ± 0.003 units (pH ~ 7.93) in 2100, while in the worst scenario (SSP5-8.5), in which CO_2 emissions double until 2050, the ocean's mean pH can decrease 0.37 ± 0.007 units (pH ~ 7.73) (Cooley et al., 2023).

Although the effects of pH are currently being assessed in experimental studies, the tested pH ranges were too broad and did not represent possible acidification scenarios, as the changes that can occur in the marine environment are more subtle. Therefore, these experiments may underestimate or overestimate the effects of the acidification process. Moreover, as far as we know, CO_2 -driven acidification has yet to be evaluated in these specific experiments, as it was utilized acid/base solutions to change the pH of the medium, which is a simpler method to carry out but has less environmental relevance.

Hence, this study aims to evaluate the availability of copper in a marine environment contaminated with photodegraded polyethylene terephthalate (PET) microplastics under two ocean acidification treatments: the moderate acidification (pH 7.93) and intense acidification scenarios estimated for 2100 (pH 7.73), with emphasis on the abiotic conditions affecting it. We hypothesize that decreasing pH would reduce PET adsorption capacity due altered carbonate speciation, thereby increasing dissolved and potentially bioavailable Cu concentration.

2 Methodology

2.1 Microplastic Preparation

The polymeric type of microplastic chosen for the experiments was polyethylene terephthalate (PET), as it is listed among the most frequently found microplastics in the ocean. The microplastics were produced by grinding transparent plastic caps in a cryogenic mill (IKA). After this process, the microplastics were passed through two sieves with mesh openings of 500 μm and 1 mm, defining the size range of the microplastics for the experiments.

To perform accelerated photodegradation, the microplastics were exposed (using a distance of 25 cm) to a medium pressure mercury lamp (400 W) which emits radiation at wavelengths of 200 to 700 nm, for 30 h, according to the procedures of da Silva et al. (2023). The plastic particles were placed in an uncovered Petri dish and mixed with a metal spatula every 2 h to ensure that the entire surfaces of the particles were exposed to radiation. The accelerated photodegradation exposure time is equivalent to environmental exposure of 174 days in the city of São Paulo (Brazil) (the calculations are present in the Supplementary Material).

2.2 Experimental Medium

The marine water utilized in the experiment was collected from Cassino Beach (Rio Grande, Rio Grande do Sul – Brazil) by a pumping system from the *Estação Marinha de Aquacultura* (EMA-FURG) (32°12'16" S, 52°10'40" W) in April 2024. The water was filtered through a sand filter and a biological filter and then subjected to ultraviolet treatment after sampling. After this process, the water was chlorinated once (10 ppm) to prevent microbial growth after the pumping of seawater from Cassino Beach. The water was maintained under constant aeration, which promotes volatilization and natural dissipation of residual chlorine. One week later, residual concentrations were measured by spectrophotometry and were below detection limits (0 mg L⁻¹), indicating complete dissipation of chlorine prior to the start of the experiments.

At this stage, water samples were collected for the analysis of total suspended solids (TSS) according to the method of (Strickland and Parsons, 1972).

Up to 20 mL of water was filtered through gravimetry on glass-fibre GFF filters (Whatman, 0.45 μm), which were then dried for approximately 24 h at 60°C and then weighed on an analytical balance (Sartorius MC1, analytic AC 210 S) with a precision of 0.0001 g to determine the final weight (AOAC, 2000).

After two months, the water was transported to the *Laboratório de Microcontaminantes Orgânicos e Ecotoxicologia Aquática* (CONECO-FURG), where the experiment was conducted. Before the next stages, water parameters were measured: The pH was 8.14 and the salinity of the marine water was 39, being later corrected to 35 through dilution with deionized water. Although the prolonged storage time of seawater may potentially alter dissolved organic matter content, particulate load, microbial activity, and the concentration metal-binding ligands, prior to its use in the experiments, the water was stored in the dark at controlled temperature conditions in order to reduce photochemical reactions and biological activity. Therefore, even though some degree of alteration in ligands cannot be entirely excluded, all experiment treatments were conducted using the same water batch, ensuring internal consistency and allowing valid comparisons among experimental conditions

2.3 Preparation of Copper Solutions

The stock Cu solution was prepared from copper (II) chloride-dihydrate (CuCl₂·2H₂O) (Dinâmica) and deionized water at a nominal concentration of 10 mg/L (acidified with 1% nitric acid). From this stock solution, 5 $\mu\text{g/L}$ solutions were prepared for the experiment by diluting seawater (experimental medium). Each solution was adjusted to the specific pH of the corresponding treatment before use. This specific Cu (II) concentration was chosen because it is considered the limit concentration acceptable for saline waters by Brazilian legislation (CONAMA resolution n.357/2005) and is close to the limit of the dissolved Cu acute concentration (4.8 $\mu\text{g/L}$) not expected to pose a significant risk to the majority of species by the United States legislation (USEPA, 2024).

2.4 Acidification System

To create an acidified condition, a CO₂ gas cylinder was used to decrease the pH of a portion of the water

for the intense acidification treatment (pH 7.7) 3 days before the experiment started to ensure that the system pH was preequilibrated. The gas was injected through a hose connected to a solenoid valve that could be opened or closed manually, according to the methods of Lopes et al. (2020). During the experiment, the pH was checked using an equipment containing a probe (Black Stone - BL981411 pH controller) (precision of ± 0.1 pH) that was located inside a beaker filled with marine water, which was placed alongside the experimental replicates inside a plastic box (Figure 1). When necessary, CO_2 was injected again into another beaker containing only seawater, where the gas equilibrated with the atmosphere contained in the box and then was absorbed by the water inside the other beakers, acidifying all the experimental units (Altafim et al., 2023; Alves et al., 2024).

For the moderate acidification treatment, the CO_2 injection system was not necessary, as the water pH naturally decreased when the salinity was corrected with the addition of deionized water. Therefore, there

is one less beaker inside each box of this pH treatment (Figure 1).

2.5 Copper Dynamics Experiment

To evaluate the effects of acidification on Cu dynamics in the marine environment, a 48-h experiment was conducted with two treatments: the moderate acidification (pH 7.9 - Cooley et al., 2023) and the intense acidification scenarios estimated for 2081-2100 (pH 7.7 - Cooley et al., 2023). The experiment was conducted in a climate-controlled room where the temperature was maintained at 20°C using an air conditioning system.

The experiments were held inside closed plastic boxes in triplicate for each pH treatment ($n=3$ for the moderate acidification treatment, $n=3$ for the intense acidification treatment) (Figure 1). To evaluate the behaviour of Cu (II) in the presence of microplastics, two glass beakers containing 250-mL of marine water were placed in all boxes and had the addition of 1 g/L

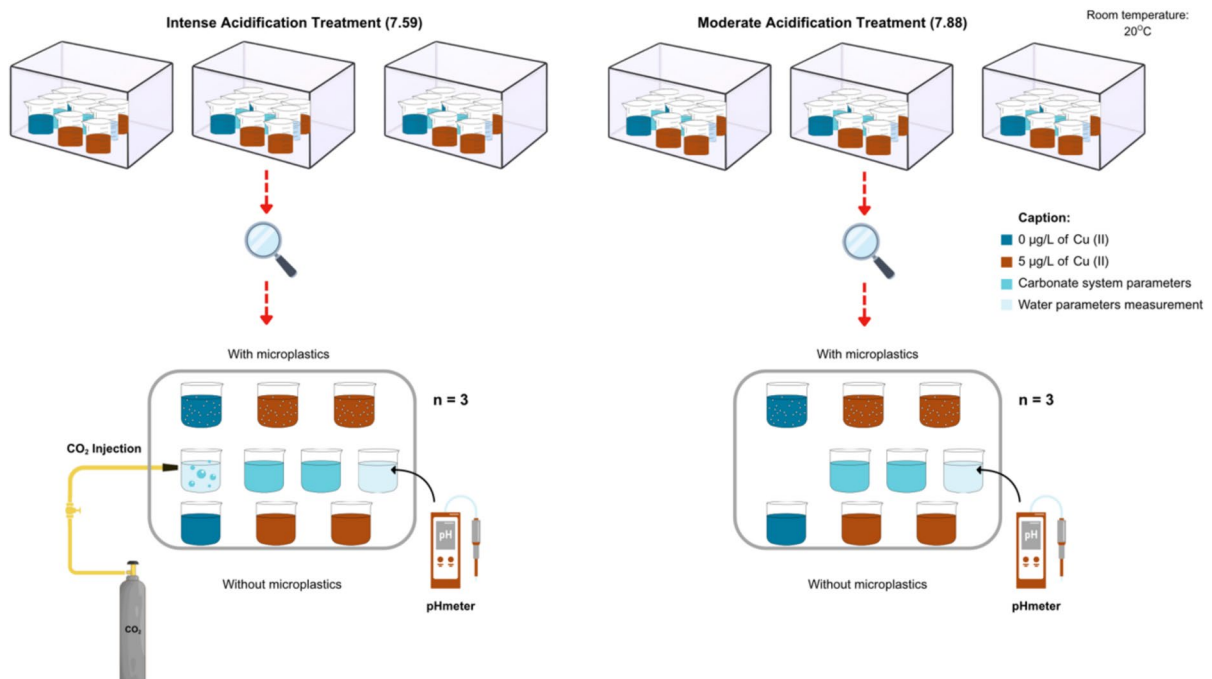


Fig. 1 The experimental design for the intense acidification treatment (pH 7.59) is shown on the left. The experimental units were placed inside boxes in triplicate, using treatments with the addition of copper (5 $\mu\text{g/L}$) and without (0 $\mu\text{g/L}$), and with the presence of microplastics (1 g/L) and without. Units were also added for the analysis of carbonate system param-

eters. In this treatment, a CO_2 injection system was used to reduce the pH of the medium. On the right is shown the experimental design of the moderate acidification treatment (pH 7.88), also containing the experimental units inside three boxes, but without the presence of the CO_2 injection system

of photodegraded PET microplastics and 5 µg/L of the Cu (II) solution. Additionally, one glass beaker containing only marine water and 1 g/L of microplastics (no copper solution) was placed inside each box to assess the behaviour of microplastics with the naturally occurring Cu in the marine water. Moreover, to evaluate the metal adsorption to the naturally occurring particles present in the marine water, two glass beakers had the addition of the Cu (II) solution (without microplastics), while another beaker containing only marine water was considered the control group. Moreover, in the intense acidification treatment, another beaker was placed inside each box to receive the injection of CO₂ when the pH decreased, as was described in the last section. Lastly, to determine the variations in the carbonate system resulting from pH changes, two 500-mL beakers containing only marine water were placed in each box.

The water inside all the glass beakers remained static during the 48 h experiment, with no circulation. Since some microplastics remained floating in the water and all its faces might not have been in contact to the Cu (II) solution, a small amount of the aqueous medium was removed with an automatic volumetric pipette and placed on the surface of the remaining floating microplastics.

Aliquots of 10 mL of water were collected at the beginning of the experiment and after 48 h of exposure from each treatment for the analyses of dissolved Cu, taking care not to collect the microplastics. All the water aliquots were filtered through syringes containing 0.45-µm filters and then acidified with nitric acid (1%). Aliquots of 250 mL of water were also collected for carbonate system analyses at the beginning and end of the experiment, according to the procedure described by Dickson et al. (2007). Water properties such as salinity (precision of 1% FS + one digit), pH (precision of ± 0.1 pH) and temperature (precision of ± 0.5°C) were measured in a separate beaker (only marine water) with a multiparameter probe (AKSO) at the beginning and end of the experiment to prevent CO₂ from escaping from the boxes throughout the experiment.

The Cu availability was estimated by taking the difference between the initial concentration of the metal and the final concentration present in the water aliquots (Zou et al., 2020).

It is important to mention that all the glassware utilized throughout the experiment was previously

decontaminated through a washing process, being firstly placed in a detergent bath (4%) for 24 h and then a nitric acid (2.5%) bath for another 24 h. Afterwards, the glassware was washed with ketone and alcohol and, finally, rinsed with distilled water.

2.6 Copper Analysis and Speciation

The previously collected water aliquots were desalinated according to Nadella et al. (2009), and the dissolved fraction of Cu (bioavailable) (0.45-µm filter) was analysed by a graphite furnace atomic absorption spectrometer (ContrAA 700), with a low quantification limit of 0.05 µg/L. The calibration curve was generated with multielement standard solution (Sigma–Aldrich) (maximum concentration: 100 µg/L or 25 µg/L, depending on the sample Cu content). Copper speciation was estimated by the geochemistry model Visual MINTEQ (V3.1) using the average dissolved Cu (II) concentrations from both pH treatments, along with the mean temperature and pH values alongside mean DIC concentrations.

2.7 Carbonate System Analysis

For carbonate system determinations, the water aliquots collected at the beginning (t=0h) and end of the experiment (t=48h) were fixed with a supersaturated solution of mercury chloride (HgCl₂) to prevent alterations in the parameters due to biological activity prior to analysis. The total alkalinity (TA) of the samples was analysed by potentiometric titration with 0.1 N hydrochloric acid (HCl) in an open-cell system, according to the method described by Dickson et al. (2007), with a precision of ±1 µmol kg⁻¹, whereas DIC was determined via a closed-cell potentiometric titration method, also according to Dickson et al. (2007), with a precision of ±3.7 µmol kg⁻¹. The results were used to calculate the partial pressure of CO₂ (pCO₂) and the concentrations of HCO₃⁻, CO₃²⁻ and CO₂ using CO₂SYS software (V3.0)(Pierrot et al. 2021), utilizing the input data of pH in total scale, the CO₂ constants k1 and k2 from Lueker, the KSO₄ source from Dickson, the KF source from Dickson and Riley, the Total Boron source from Uppstrom and the Equation of State EOS-80.

2.8 Statistical Analysis

The normality of the data was assessed using the Shapiro–Wilk test, whereas the homoscedasticity was evaluated with Levene’s test. When necessary, a Box–Cox test was applied to normalize the data. An orthogonal two-way ANOVA was then conducted to test for differences between the pH treatments, the presence or absence of microplastics, and their interaction. Tukey’s post hoc test was subsequently applied to explore pairwise differences further. Owing to the exploratory nature of this study and the complex interactions among pH, microplastics, and copper, we adopted a significance threshold of $\alpha = 0.1$. This approach aims to reduce the risk of Type II errors, i.e. failing to detect meaningful trends that may have important ecological implications. Although this threshold increases the likelihood of Type I errors, it allows for a more cautious interpretation of marginally significant results (Underwood, 2011) and ensures that potential biological effects are not overlooked in this preliminary investigation. All the statistical analyses were performed using R (R

Core Team). Additional information on the results is provided in the supplementary material (Table S2).

3 Results

3.1 Water Quality Parameters

Both treatments resulted in a mean water temperature of 20°C and a salinity of approximately 35. However, the mean pH values in both treatments (moderate and intense acidification) slightly were lower than the target levels defined for the experimental scenarios, especially in the intense acidification treatment (Table 1).

3.2 Carbonate System Parameters

In the intense acidification treatment, while the concentrations of $p\text{CO}_2$, CO_2 , HCO_3^- , TA and DIC increased, the concentration of CO_3^{2-} decreased compared to the moderate acidification treatment (Table 2, Table 3).

Table 1 Mean ($\pm$ standard errors) of the experimental parameters for each treatment. TSS = total suspended solids

Treatments		Water Temperature (°C)	pH	Salinity	TSS (mg/L)
Moderate acidification (pH 7.93)	Initial n=3	20.0 $\pm$ 0	7.960 $\pm$ 0	34.800 $\pm$ 0	-
	Final n=3	21.1 $\pm$ 0.115	7.800 $\pm$ 0.032	35.100 $\pm$ 0.032	-
	Average	20.55 $\pm$ 0.251	7.880 $\pm$ 0.041	34.950 $\pm$ 0.041	2.4
Intense acidification (pH 7.73)	Initial n=3	19.63 $\pm$ 0.033	7.500 $\pm$ 0	35.033 $\pm$ 0	-
	Final n=3	20.77 $\pm$ 0.033	7.677 $\pm$ 0.012	35.367 $\pm$ 0.012	-
	Average	20.20 $\pm$ 0.254	7.588 $\pm$ 0.040	35.200 $\pm$ 0.040	2.4
Difference between treatments		0.35	0.292	0.25	-

Table 2 Mean ($\pm$ standard error) total alkalinity (TA), dissolved inorganic carbon (DIC) and pH for each treatment. The symbol (*) indicates that the parameters were measured in a single sampling

Treatments	TA ($\mu\text{mol kg}^{-1}$)	DIC ($\mu\text{mol kg}^{-1}$)	pH (total scale)
Initial moderate acidification (pH 7.88) n=3	2381 $\pm$ 3	2241 $\pm$ 3	7.75 $\pm$ 0
Final moderate acidification (pH 7.88) n=3	2373 $\pm$ 2	2272 $\pm$ 10	7.66 $\pm$ 0.02
Average	2377 $\pm$ 3	2257 $\pm$ 8	7.706 $\pm$ 0.023
Initial intense acidification (pH 7.59) n=1	2410 *	2440 *	7.435 *
Final intense acidification (pH 7.59) n=1	2408 *	2353 *	7.568 *
Average	2409 $\pm$ 1	2396 $\pm$ 43	7.50 $\pm$ 0.07
Difference between treatments	32	139	0.206

Table 3 Mean ($\pm$ standard error) carbonate system parameters calculated by the software CO2sys

	pCO ₂ (μ atm)	HCO ₃ ⁻ (μ mol kg ⁻¹)	CO ₃ ²⁻ (μ mol kg ⁻¹)	CO ₂ (μ mol kg ⁻¹)
Moderate acidification (pH 7.88) n=6	1024.33 $\pm$ 122.62	2133.44 $\pm$ 14.43	99.76 $\pm$ 7.67	32.54 $\pm$ 3.39
Intense acidification (pH 7.59) n=2	1731.07 $\pm$ 278.97	2248.49 $\pm$ 27.38	65.9 $\pm$ 10.88	55.68 $\pm$ 9.90
Difference between treatments	706.64	115.05	33.86	23.14

3.3 Copper Dynamics in Different pH Environments

The initial and final concentrations of dissolved Cu (II) for all the treatments are provided below (Table 4). In the moderate acidification treatment, the treatment without microplastics but with the addition of Cu (II) had a decrease in the metal concentration after 48 h. In the treatment with the presence of microplastics, the control treatment had an increase of Cu (II) concentration at the end of the experiment, while the opposite happened to the copper treatment. In the case of the intense acidification treatment without microplastics, both control and copper treatments had an increase of Cu (II) concentrations after 48 h. In the treatment with the presence of microplastics, the control treatment had an increase of Cu (II) concentration, while the copper treatment had a decrease. Although the increase of Cu (II) in both control treatments is unexpected, it is most plausibly attributed to pH-driven redistribution of copper between complexed/particulate and dissolved phases rather than contamination.

After obtaining the results of Cu (II) concentrations, it was decided to group together the control (0 μ g/L) and copper (5 μ g/L) treatments to evaluate the

trends in Cu availability across the pH and microplastic treatments (Figure 2). As previously described, the dynamics of dissolved Cu (II) and its consequent availability in the water were estimated as the difference between the initial and final concentrations of the metal in the water. In the intense acidification treatment, this difference was negative, particularly in the absence of microplastics, suggesting a tendency for higher copper concentrations than those in the treatment with microplastics (Figure 2). These results indicate that, on average, the final dissolved Cu (II) concentration was higher than the average initial concentration, implying that more dissolved Cu (II) was available in the system at the end of the experiment under conditions of intense acidification. Conversely, in the moderate acidification treatment, the difference between the mean initial and final concentrations was positive (Figure 2) regardless of the presence of microplastics or not, indicating that less dissolved Cu (II) was available in the water after 48 h.

Considering the significance threshold of $\alpha = 0.1$, both the pH 7.88 and the pH 7.59 treatments were significantly different ($p = 0.063$), while there was no significant difference between the treatments with a presence or absence of microplastics ($p = 0.612$).

Table 4 Mean ($\pm$ standard error) initial and final dissolved Cu (II) concentrations (μ g/L) of all treatments

Treatments	Cu treatments	Moderate acidification (7.88)		Intense acidification (7.59)	
		Initial	Final	Initial	Final
Without microplastics	Control n=3	0.883 $\pm$ 0.238	0.883 $\pm$ 0.238	0.883 $\pm$ 0.238	1.641 $\pm$ 0.367
	Copper n=6	3.936 $\pm$ 1.146	2.359 $\pm$ 0.858	2.514 $\pm$ 0.395	3.428 $\pm$ 1.003
With microplastics	Control n=3	0.883 $\pm$ 0.238	1.698 $\pm$ 0.863	0.883 $\pm$ 0.238	1.702 $\pm$ 0.569
	Copper n=6	3.936 $\pm$ 1.146	2.011 $\pm$ 0.465	2.514 $\pm$ 0.395	2.357 $\pm$ 0.140

Fig. 2 Difference between initial and final dissolved copper (Cu) concentrations in the water samples of the moderate acidification treatment (7.88) and the intense acidification treatment (7.59), both with and without microplastics (MPs). The bars represent the mean concentration and their respective standard errors. The chosen significance test was an orthogonal two-way ANOVA, with a significance threshold of $\alpha = 0.1$



(Figure 2). For the interaction between these two factors (pH and microplastics treatment), there was also no significant difference ($p = 0.411$).

3.4 Copper Speciation

In the moderate acidification treatment, at the end of the experiment, the majority of Cu was in the form of $\text{CuCO}_3(\text{aq})$ (90.24%), followed by CuOH^+ (4.51%) and Cu^{2+} (2.6%). For the intense acidification treatment, the most common Cu species was also $\text{CuCO}_3(\text{aq})$ (88.92%), followed by Cu^{2+} (4.99%) and CuOH^+ (4.36%). More information about the Cu speciation results can be found in the supplementary material (Table S1).

A comparison of both treatments revealed that the speciation changed as the pH decreased: the percentages of $\text{CuCO}_3(\text{aq})$ and CuOH^+ decreased in the intense acidification treatment, whereas the percentage of free Cu (II) (Cu^{2+}) increased.

4 Discussion

In this study, the availability of dissolved Cu (II) in a simulated marine environment in the presence of PET microplastics was tested under two scenarios with different pH levels: one representing a moderate acidification at the surface of the ocean (pH 7.88) and the other representing an intense acidification scenario (pH 7.59), both estimated for 2100. Considering the

trends in Cu (II) availability (Figure 2), it was discovered that the reduction in pH increased the availability of free Cu (II) ions in the experimental setting, but the presence of microplastics did not have a notable role in its concentration. The intense acidification scenario also resulted in increases in HCO_3^- , $p\text{CO}_2$, TA and DIC and a decrease in the concentration of CO_3^{2-} . The chemical changes that occurred in both pH treatments are shown in Figure 3.

4.1 Copper Availability at the Moderate Acidification Treatment

As expected, under moderate acidification conditions (7.88), the carbonate system was dominated by bicarbonate ions (HCO_3^-) but also presented a relatively high concentration of carbonate ions (CO_3^{2-}) (Table 3), which constitute the primary inorganic ligand of Cu (II) in the ocean (Millero et al., 2009). The measured parameters of the carbonate system (Tables 2 and 3) were consistent with those reported by Alves et al. (2024), who observed similar values at pH 7.7.

Both treatments with and without microplastics showed a similar trend of decrease in the dissolved Cu (II) concentrations in the water after 48 hours of the experiment (Figure 2). The absence of a statistically significant difference between these treatments suggests that, under the tested pH conditions, microplastics did not play a notable role in the adsorption of Cu (II). Even though the concentration

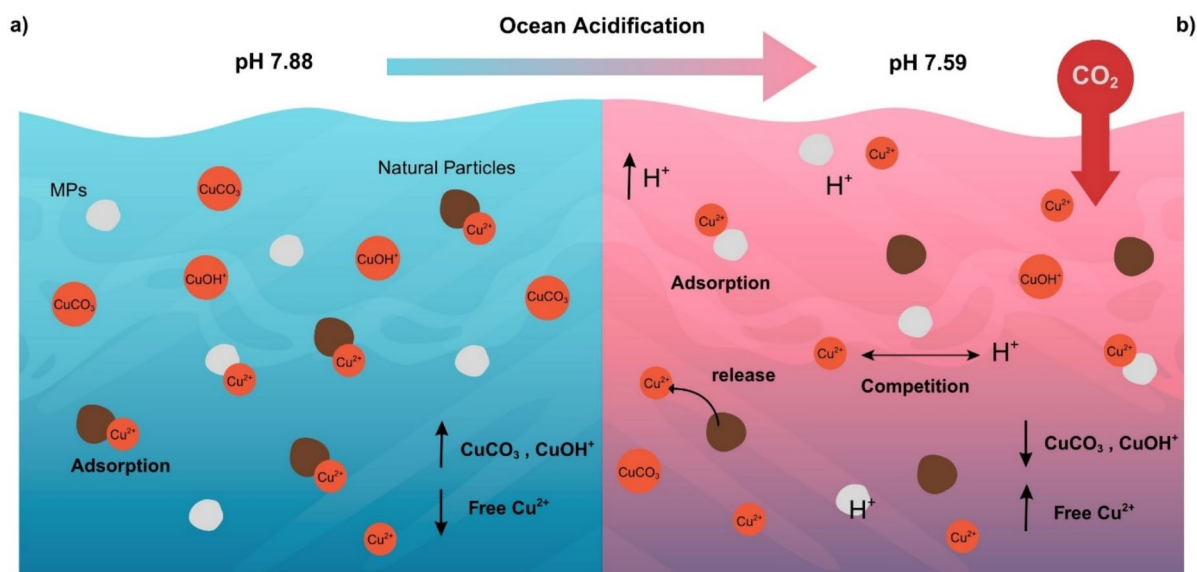


Fig. 3 Resume of the chemical processes in both pH treatments. A) In moderate acidification treatment, dissolved Cu was adsorbed on natural particles rather than on microplastics, which had a limited adsorption capacity. At this pH, there is a higher occurrence of complexed Cu (CuCO_3 and CuOH^+) than free ionic Cu (Cu^{2+}). B) In the intense acidification treatment

(pH 7.59), a release of Cu (II) previously adsorbed on natural particles likely occurred. Microplastics had some adsorption capacity, but not sufficient to decrease dissolved Cu concentrations, since H^+ and Cu^{2+} have to compete for microplastics' sorption sites. At a lower pH, CuCO_3 and CuOH^+ concentrations decreased, while Cu^{2+} was favoured

of microplastics (1 g/L) chosen for this experiment is environmentally relevant, as it is within the range found in estuaries (Zhang et al., 2025), one possible explanation is that it was insufficient to effectively adsorb the available Cu (II) ions. Wang et al. (2020) demonstrated that the Cu and Zn adsorption efficiencies in the HNO_3 medium increased with the dosage of microplastics, with the lowest adsorption observed at 1 g/L and a significant increase occurring at 2 g/L and above, where the adsorption plateaued.

Although the sizes of microplastics used in this study (500 μm to 1 mm) reflect environmentally relevant conditions and align with the size distributions reported in the Atlantic Ocean (Enders et al., 2015), the broad particle size range may affect the adsorption efficiency. Several studies have shown that smaller microplastics possess higher adsorption capacities. For example, Wang et al. (2019) reported that HDPE particles in the 100–154 μm range exhibited significantly greater Cd adsorption than particles measuring 0.6–1 mm and 1–2 mm. Similar findings were presented by Han et al. (2021), where PE, PP, and PET particles < 0.9 mm exhibited greater Cu (II) adsorption than did particles in the 0.9–2 mm and

2–5 mm ranges. Accordingly, smaller particles have greater specific surface areas, more adsorption sites and, therefore, greater capacities for metal adsorption (Wang et al., 2019). Hence, microplastics at the upper end of the tested size range may have experienced a reduced potential for adsorption.

Additionally, the different sizes of the microplastics may make them not chemically uniform, as there are variations in surface morphology, degree of photodegradation, and distribution, consequently leading to different adsorption efficiencies and higher dispersion among replicates (Table 4) (Han et al., 2021).

Despite the relatively low concentration of total suspended solids (2.4 mg/L), their potential contribution to Cu (II) adsorption should not be disregarded. (Chen et al., 2021) demonstrated that natural particles such as sediments possess significantly higher copper-binding capacities than microplastics (PE and PS), whether they are pristine or weathered. This was attributed to their rougher surfaces, greater surface areas, and presence of oxygen-containing functional groups on the natural particles. Accordingly, Guan et al., (2020) reported adsorption of greater amounts of Cu (II) and other metals onto freshwater suspended

solids and surface sediment than onto expanded PS particles. Furthermore, compared with virgin microplastics, microplastics covered by biofilms presented increased metal adsorption.

The adsorption mechanisms likely depend on the types of polymers and the chemical characteristics of the contaminants (Torres et al., 2021). Physical adsorption mechanisms, which include electrostatic interactions, are considered one of the major mechanisms for microplastics with simpler structures (Gao et al., 2021). However, in the case of PET microplastics, several studies suggest that chemical adsorption mechanisms occur instead (Bernardino et al, 2023; Godoy et al., 2019; Wang et al, 2020). Therefore, under the conditions of the present study, chemical adsorption may have been the major mechanism operating in the interaction between PET microplastics and Cu (II) at the tested pH.

4.2 Copper Availability in an Environment Under intense Acidification

Under conditions of intense acidification (pH 7.59), the carbonate system responded as expected to the increase in CO₂ availability, with a decrease in both pH and CO₃²⁻ concentration, accompanied by increased HCO₃⁻ and DIC concentration (Tables 2 and 3). These changes confirmed that the carbonate system was effectively altered in response to acidification. These results were consistent with those obtained by Balbela et al. (2024) and Alves et al. (2024), who also simulated CO₂-driven acidification in similar experimental setups.

Regarding the increase in TA concentration (Table 2), it is likely due to the salinity correction with deionized water, since TA and DIC are salinity dependent. After the CO₂-injection in this treatment, DIC increased and carbonate speciation shifted, while TA remained relatively stable within analytical uncertainty.

With respect to the Cu dynamics under these conditions, a higher concentration of dissolved Cu (II) was observed at the end of the experiment than at the beginning (Figure 2). This suggests that part of the Cu (II) that was previously adsorbed on natural particles was released into the aqueous phase because of the lower pH.

Although no statistically significant difference was detected between the treatments with and without

microplastics, distinct trends were observed. Both treatments resulted in increases in the concentration of dissolved Cu (II) after 48 hours (Figure 2). However, the treatment with microplastics consistently resulted in a lower dissolved Cu (II) concentration, suggesting a modest capacity of the microplastics to adsorb Cu, although this capacity was not sufficient to completely reduce the concentration of the dissolved metal.

The adsorption of metals onto microplastics is influenced by multiple factors, including the polymeric composition of the plastic particles. With respect to PET-based microplastics, the results in the literature are inconsistent. For example, Godoy et al. (2019) investigated metal adsorption onto microplastics of different polymer types in natural seawater (pH 7.72) and reported that PET microplastics had the lowest adsorption capacity for most metals, including Cu, potentially because of their low pore volumes and the presence of functional fillers, which are additives that protect the plastic against ageing and biofouling. On the other hand, the photodegraded PET and PE microplastics exhibited significantly greater Cu (II) adsorption than PP did (Han et al., 2021), indicating that weathering processes can enhance the surface reactivity and metal binding affinity through morphological changes. Bernardino et al. (2023) reported that PET microplastics had Cu (II) removal efficiencies ranging from 5% to 40% in acidic water (pH 4.0). Despite the relatively low surface areas of the PET particles, their polymeric composition was highly susceptible to adsorption due to hydrogen bonding, dipole–dipole interactions, van der Waals interactions, and cavity formation. Therefore, the adsorption capacity of PET-based microplastics may be affected by some factors, such as their chemical properties, the presence of additives, morphological changes induced by weathering and the chosen metal.

A possible explanation for the reduced adsorption capacity of microplastics in the intense acidification treatment is that, at a lower pH level, a high concentration of H⁺ ions may compete with Cu²⁺ ions for the sorption sites on the surfaces of microplastics (Wang et al., 2022). Similar results were reported in a study conducted by Bao et al. (2022), where aged microplastics (PET and PA) were exposed to a Cu solution in artificial seawater, and their adsorption capacities were evaluated across a pH range of 4 to 11. The adsorption capacity increased progressively

with increasing pH from 5 to 10, demonstrating that adsorption was hampered at lower pH values.

Ocean acidification can modify the chemical properties of microplastics and therefore affect their adsorption behaviour. As the pH increases, surface functional groups become deprotonated, increasing the electronegativity of microplastics and consequently benefiting the adsorption process (Zou et al., 2020). Therefore, it is possible that the opposite effect occurred in the present experiment: the decrease in pH may have reduced the electronegativity of the PET microplastics and, consequently, limited their capacity to adsorb positively charged ions such as Cu^{2+} . On the other hand, ocean acidification may also enhance metal adsorption, since CO_2 -driven acidification has been shown to increase the photooxidation of certain microplastic polymers, such as PP and EVA, while EPS did not exhibit any degradation changes (Balbela et al., 2024). Although the effect of pH on the photo-degradation of PET microplastics was not evaluated in the present study, it remains possible that this polymer type underwent further degradation in the intense acidification treatment.

As previously discussed, microplastic adsorption mechanisms are often governed by electrostatic interactions. However, numerous studies have demonstrated that Cu (II) adsorption onto PET microplastics is primarily chemically driven. Another important factor influencing these mechanisms is the environmental pH. Under alkaline conditions, adsorption may occur via electrostatic interactions, surface microprecipitation, surface complexation and metal precipitation (Bao et al., 2022). Conversely, acidic conditions tend to favour surface complexation, electrostatic interactions, competitive adsorption and cation exchange (Bao et al., 2022). Although the present study was conducted under realistic acidified conditions, the pH remains close to neutral (7.59), suggesting that the dominant mechanisms may still resemble those typically observed under alkaline conditions. Nevertheless, if extreme ocean acidification reduces the seawater pH below 7, the dominant adsorption mechanisms may change, and the Cu (II) behaviour may differ.

Changes in the carbonate system affected the inorganic speciation of Cu (II) in the water. Under typical conditions, Cu (II) is strongly complexed with carbonate; as the pH decreases, the CO_3^{2-} concentration decreases, leading to an increase in the free ionic form of Cu (Millero et al., 2009). This is confirmed

by the speciation results (Table S1), which revealed increases in the concentrations of Cu^{2+} ions and bicarbonate complexes (CuHCO_3^+) under conditions of intense acidification, whereas the concentrations of species associated with carbonate and hydroxide (OH^-) decreased.

Since the concentration of dissolved Cu (II) in the water after 48 h was higher than the initial Cu (II) concentration that was added at the beginning of the experiment, it could be inferred that, in this specific scenario, ocean acidification increased the environmental availability of the metal, especially its free ionic form. Moreover, under the tested conditions of intense acidification, the potential of the microplastics to act as vectors of Cu^{2+} was not sufficient to completely decrease its concentration, increasing the bioavailability of the metallic cations.

4.3 Environmental Consequences

As the percentage of Cu^{+2} under intense acidification conditions was greater than under the moderate acidification conditions, it is plausible that, under this specific reduced pH scenario, this metal could become more bioavailable in water and have negative effects on marine biota since Cu (II) is more toxic in its free ionic form than when it is bound in strong complexes (Millero et al., 2009). CO_2 -driven Ocean acidification alone has negative effects on the environment, such as the dissolution of the aragonite shells of Pteropods (Orr et al., 2005). However, as this process occurs concomitantly with other environmental stressors, the combination of ocean acidification and Cu contamination has become a subject of interest in bioassays, which yield varied results according to the studied species. Some examples include DNA damage in mussels and sea urchins (Lewis et al., 2016); impaired physiology, growth and development of jellyfish (Wang et al., 2025); oxidative stress in juvenile flounder (Xiao et al., 2023); DNA damage, diminished larval survival and sperm mobility in polychaeta (Campbell et al., 2014); decreased growth and food consumption, and histological damage, among others, in cephalopods (Zheng et al., 2023). These studies thus highlight that, in intensely acidified environments, marine organisms, especially sensitive species, are more susceptible to physiological stress, not only because of changes in water chemistry but also because of increased Cu toxicity. In the presence of microplastics, these particles

may tend to decrease some of the dissolved Cu (II) but are not sufficient to mitigate the Cu^{2+} concentration and completely reduce its toxicity.

4.4 Limitations of the Experiment

As was previously stated, the obtained pH for both treatments [moderate (7.88) and intense acidification (7.59)] were slightly lower than the stipulated values (7.93 and 7.73). Despite that, the treatments had a difference of 0.3 units, which clearly chemically differentiates both environments. For the moderate acidification treatment, one possible explanation for the lower value is that the handling of seawater to the experimental beakers may have caused this decrease in pH, as well as the addition of deionized water to lower the seawater salinity, which could have affected the marine water's buffer capacity. As for the intense acidification treatment, the CO_2 injection system was not as precise as expected, and the introduction of a small quantity of gas slowly lowered the water pH below the chosen value, making it difficult to adjust.

Another limiting factor was the absence of measurement of organic matter in natural marine water, as it forms strong complexes with Cu (II) in seawater (Millero et al., 2009) and may difficult the adsorption onto microplastics. On the other hand, dissolved organic matter may facilitate the interaction between metal ions and microplastics (Gao et al., 2021), but they also may have a stronger binding affinity for copper than virgin microplastics, primarily due to their chemically heterogeneous surfaces and the presence of diverse functional groups capable of forming stable metal complexes (Chen et al., 2021). Hence, the complex interaction between microplastics, metal ions and organic matter, especially in a context of increased microplastics marine pollution and climate change, should be further studied.

5 Conclusion and Future Recommendations

In conclusion, at the moderate acidification treatment (7.88), the concentration of dissolved Cu (II) decreased after 48 h, primarily due to adsorption onto natural particles rather than onto microplastics, which exhibited a limited capacity to adsorb Cu^{2+} under the experimental conditions. Conversely, in the intense acidification scenario (7.59), the concentration of dissolved Cu (II)

increased, likely due to the release of Cu previously adsorbed on natural particles and also the reduced adsorption capacity of the microplastics, as Cu^{2+} competes with the increased H^+ concentrations for the sorption sites on the surfaces of microplastics. These results indicate that under intense pH reduction, the capacity of microplastics to act as vectors of Cu could be insufficient, potentially increasing the concentration of free ionic-Cu and increasing its availability.

It is important to further evaluate the effects of ocean acidification on metal adsorption, particularly across various concentrations, polymer types, and size ranges of microplastics, as well as with different metal ions, since these factors may significantly influence the adsorption capacity. Additionally, potential metal ion desorption in acidified environments should be considered, as this process could increase the availability of dissolved Cu in water, thereby exacerbating its toxicity to biota. Finally, it is important to emphasize that ocean acidification does not occur in isolation but rather concomitantly with other anthropogenic disturbances, such as ocean warming and deoxygenation. Therefore, multistressor experiments are critically important, especially those that examine the long-term consequences for both the environment and the communities it supports.

Acknowledgments The authors would like to thank Estação Marinha de Aquicultura (EMA) for the supplying of seawater and Laboratório de Estudos dos Oceanos e Clima (LEOC) for water chemical analysis, both laboratories present at the Federal University of Rio Grande (FURG) in Rio Grande, RS, Brazil. We thank the resources provided by CAPES Foundation to support the Graduate Program in Oceanology (FURG). We would like to thank Editage (www.editage.com) for English language editing.

Author Contributions Conceptualization: Luísa Silva Poersch, Grasiela Lopes Leães Pinho, Sanye Soroldoni, **; Methodology: Luísa Silva Poersch, Grasiela Lopes Leães Pinho, Sanye Soroldoni, Camila De Martinez Gaspar Martins, Fernanda Chaves Lopes **, ** Marco Tadeu Grassi **, Formal analysis: Luísa Silva Poersch, Maurício Garcia de Camargo, Sanye Soroldoni **, Investigation: Luísa Silva Poersch, Fernanda Chaves Lopes, Brenno Januário Silva, Sanye Soroldoni, Camila De Martinez Gaspar Martins **, Data Curation: Luísa Silva Poersch, Grasiela Lopes Leães Pinho, Sanye Soroldoni **, ** Camila De Martinez Gaspar Martins, Fernanda Chaves Lopes, Brenno Januário Silva, Maurício Garcia de Camargo; Writing - original draft preparation: Luísa Silva Poersch **, Writing - review and editing: Sanye Soroldoni, Fernanda Chaves Lopes, Camila De Martinez Gaspar Martins, Marco Tadeu Grassi, Brenno Januário Silva, Maurício Garcia de Camargo, Rodrigo Kerr, Grasiela Lopes Leães Pinho **, Funding acquisition: Grasiela Lopes Leães Pinho **, ** Sanye

Soroldoni **: Resources: Grasiela Lopes Leães Pinho, Sanye Soroldoni, Camila De Martinez Gaspar Martins **, ** Rodrigo Kerr **, Supervision: Grasiela Lopes Leães Pinho **, ** Sanye Soroldoni; Project administration: Grasiela Lopes Leães Pinho, Sanye Soroldoni.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). This research was supported by Fundação de Amparo à Pesquisa do Estado do Rio Grande do Sul (FAPERGS) (TO 18/2551-0000528-2) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) (425520/2018-8, 435612/2018-2, 406391/2021-1 and 409084/2023-9). Sanye Soroldoni is a post-doctoral fellow (PDJ no 150164/2023-8), and Grasiela L. L. Pinho a research fellow (PQ2 no 307700/2022-4) of CNPq.

Data Availability All data supporting the findings of this study are available within the manuscript and Supplementary Information.

Declarations

Ethics Approval Not applicable.

Consent for Publication On behalf of all the authors, I indicate our consent to the publication of this manuscript.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Ahechti, M., Benomar, M., El Alami, M., & Mendiguchía, C. (2022). Metal adsorption by microplastics in aquatic environments under controlled conditions: Exposure time, pH and salinity. *International Journal of Environmental Analytical Chemistry*, 102, 1118–1125. <https://doi.org/10.1080/03067319.2020.1733546>
- Altafim, G. L., Alves, A. V., Trevizani, T. H., Figueira, R. C. L., Gallucci, F., & Choueri, R. B. (2023). Ocean warming and CO₂-driven acidification can alter the toxicity of metal-contaminated sediments to the meiofauna community. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2023.163687>
- Alves, A. V., Kureki, R. K., Trevizani, T. H., Figueira, R. C. L., & Choueri, R. B. (2024). Effects of metals in sediment under acidification and temperature rise scenarios on reproduction of the copepod *Nitokra* sp. *Marine Pollution Bulletin*. <https://doi.org/10.1016/j.marpolbul.2024.117125>
- Andrady, A. L. (2017). The plastic in microplastics: A review. *Marine Pollution Bulletin*, 119(1), 12–22. <https://doi.org/10.1016/j.marpolbul.2017.01.082>
- AOAC. (2000). *Official Methods of analysis* (17th Edition). The Association of Official Analytical Chemists.
- Balbela, C. C., Soroldoni, S., Fernandes, A. N., de Camargo, M. G., Kessler, F., & Pinho, G. L. L. (2024). Assessing the impact of simulated ocean acidification on the photodegradation of selected microplastics. *Marine Pollution Bulletin*. <https://doi.org/10.1016/j.marpolbul.2024.116854>
- Bao, R., Fu, D., Fan, Z., Peng, X., & Peng, L. (2022). Aging of microplastics and their role as vector for copper in aqueous solution. *Gondwana Research*, 108, 81–90. <https://doi.org/10.1016/j.gr.2021.12.002>
- Barnes, D. K. A., Galgani, F., Thompson, R. C., & Barlaz, M. (2009). Accumulation and fragmentation of plastic debris in global environments. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364, 1985–1998. <https://doi.org/10.1098/rstb.2008.0205>
- Bernardino, C. A. R., Mahler, C. F., Fernandes, J. O., Lopes, C. S. C., Braz, B. F., Archanjo, B. S., Santelli, R. E., & Cincotto, F. H. (2023). Evaluation of microplastic contamination by metals in a controlled environment: A risk to be considered. *Environmental Monitoring and Assessment*, 195, Article 691. <https://doi.org/10.1007/s10661-023-11389-1>
- Campbell, A. L., Mangan, S., Ellis, R. P., & Lewis, C. (2014). Ocean acidification increases copper toxicity to the early life history stages of the polychaete *arenicola marina* in artificial seawater. *Environmental Science & Technology*, 48, 9745–9753. <https://doi.org/10.1021/es502739m>
- Cao, Y., Zhao, M., Ma, X., Song, Y., Zuo, S., Li, H., & Deng, W. (2021). A critical review on the interactions of microplastics with heavy metals: Mechanism and their combined effect on organisms and humans. *Science of the Total Environment*, 788, Article 147620. <https://doi.org/10.1016/j.scitotenv.2021.147620>
- Carpenter, E. J., & Smith, K. L., Jr. (1972). Plastics on the Sargasso Sea surface. *Science*, 175(4027), 1240–1241. <https://doi.org/10.1126/science.175.4027.1240>
- Chen, C. C., Zhu, X., Xu, H., Chen, F., Ma, J., & Pan, K. (2021). Copper adsorption to microplastics and natural particles in seawater: A comparison of kinetics, isotherms, and bioavailability. *Environmental Science & Technology*, 55, 13923–13931. <https://doi.org/10.1021/acs.est.1c04278>
- CONSELHO NACIONAL DO MEIO AMBIENTE - CONAMA (2005). RESOLUÇÃO Nº 357, DE 17 DE MARÇO DE 2005. Available at: <https://conama.mma.gov.br/>

- option=com_sisconama&task=arquivo.download&id=450. Accessed 2 May 2025.
- Cooley, S., Schoeman, D., Bopp, L., Boyd, P., Donner, S., Ghebrehiwet, D.Y., Ito, S.-I., Kiessling, W., Martinetto, P., Ojea, E., Racoult, M.-F., Rost, B., Kern-Mauritzen, M. (2023). *Oceans and Coastal Ecosystems and Their Services*, in: *Climate Change 2022 – Impacts, Adaptation and Vulnerability*. Cambridge University Press, 379–550. <https://doi.org/10.1017/9781009325844.005>
- da Silva, L. F., Soroldoni, S., Kessler, F., Fernandes, A. N., & Pinho, G. L. L. (2023). Influence of UV exposure time and simulated marine environment on different microplastic degradation. *Environmental Science and Pollution Research International*, 30, 121450–121464. <https://doi.org/10.1007/s11356-023-30925-7>
- Dickson, A. G., Sabine, C. L., & Christian, J. R. (Eds.). (2007). Guide to best practices for ocean CO₂ measurements (PICES Special Publication 3). North Pacific Marine Science Organization. 191 pp. https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/Handbook_2007/Guide_all_in_one.pdf
- Doney, S. C., Fabry, V. J., Feely, R. A., & Kleypas, J. A. (2009). Ocean acidification: The other CO₂ problem. *Annual Review of Marine Science*, 1, 169–192. <https://doi.org/10.1146/annurev.marine.010908.163834>
- Enamul Kabir, A. H. M., Sekine, M., Imai, T., & Yamamoto, K. (2020). Microplastics pollution in the Seto Inland Sea and Sea of Japan surrounded Yamaguchi prefecture areas, Japan: Abundance, characterization and distribution, and potential occurrences. *Journal of Water and Environment Technology*, 18, 175–194. <https://doi.org/10.2965/jwet.19-127>
- Enders, K., Lenz, R., Stedmon, C. A., & Nielsen, T. G. (2015). Abundance, size and polymer composition of marine microplastics $\geq 10 \mu\text{m}$ in the Atlantic Ocean and their modelled vertical distribution. *Marine Pollution Bulletin*, 100, 70–81. <https://doi.org/10.1016/j.marpolbul.2015.09.027>
- Gao, X., Hassan, I., Peng, Y., Huo, S., Ling, L. (2021). Behaviors and influencing factors of the heavy metals adsorption onto microplastics: A review. *J Clean Prod* 319. <https://doi.org/10.1016/j.jclepro.2021.128777>
- Gattuso, J. P., & Hansson, L. (2011). Ocean Acidification. *Oxford University Press*. <https://doi.org/10.1093/oso/9780199591091.001.0001>
- Gattuso, J. P., & Lavigne, H. (2009). Technical note: Approaches and software tools to investigate the impact of ocean acidification. *Biogeosciences*, 6, 2121–2133. <https://doi.org/10.5194/bg-6-2121-2009>
- Geyer, R., Jambeck, J. R., & Law, K. L. (2017). Production, use, and fate of all plastics ever made. *Science Advances*. <https://doi.org/10.1126/sciadv.1700782>
- Godoy, V., Blázquez, G., Calero, M., Quesada, L., Martín-Lara, M.A. (2019). The potential of microplastics as carriers of metals. *Environmental Pollution* 255. <https://doi.org/10.1016/j.envpol.2019.113363>
- Guan, J., Qi, K., Wang, J., Wang, W., Wang, Z., Lu, N., Qu, J. (2020). Microplastics as an emerging anthropogenic vector of trace metals in freshwater: Significance of biofilms and comparison with natural substrates. *Water Res* 184. <https://doi.org/10.1016/j.watres.2020.116205>
- Guo, X., & Wang, J. (2019). The chemical behaviors of microplastics in marine environment: A review. *Marine Pollution Bulletin*, 142, 1–14. <https://doi.org/10.1016/j.marpolbul.2019.03.019>
- Hale, R. C., Seeley, M. E., La Guardia, M. J., Mai, L., & Zeng, E. Y. (2020). A global perspective on microplastics. *Journal of Geophysical Research: Oceans*. <https://doi.org/10.1029/2018JC014719>
- Han, X., Wang, S., Yu, X., Vogt, R.D., Feng, J., Zhai, L., Ma, W., Zhu, L., Lu, X. (2021). Kinetics and Size Effects on Adsorption of Cu(II), Cr(III), and Pb(II) Onto Polyethylene, Polypropylene, and Polyethylene Terephthalate Microplastic Particles. *Front Mar Sci* 8. <https://doi.org/10.3389/fmars.2021.785146>
- Hanvey, J.S., Lewis, P.J., Lavers, J.L., Crosbie, N.D., Pozo, K., Clarke, B.O. (2017). A review of analytical techniques for quantifying microplastics in sediments. *Analytical Methods* 9. <https://doi.org/10.1039/c6ay02707e>
- Harmon, S.M. (2018). *The effects of microplastic pollution on aquatic organisms*, in: *Microplastic Contamination in Aquatic Environments: An Emerging Matter of Environmental Urgency*. Elsevier, 249–270. <https://doi.org/10.1016/B978-0-12-813747-5.00008-4>
- Lewis, C., Ellis, R.P., Vernon, E., Elliot, K., Newbatt, S., Wilson, R.W. (2016). Ocean acidification increases copper toxicity differentially in two key marine invertebrates with distinct acid-base responses. *Sci Rep* 6. <https://doi.org/10.1038/srep21554>
- Li, X., Mei, Q., Chen, L., Zhang, H., Dong, B., Dai, X., He, C., & Zhou, J. (2019). Enhancement in adsorption potential of microplastics in sewage sludge for metal pollutants after the wastewater treatment process. *Water Research*, 157, 228–237. <https://doi.org/10.1016/j.watres.2019.03.069>
- Liu, S., Shi, J., Wang, J., Dai, Y., Li, H., Li, J., Liu, X., Chen, X., Wang, Z., Zhang, P. (2021). Interactions Between Microplastics and Heavy Metals in Aquatic Environments: A Review. *Front Microbiol* 12. <https://doi.org/10.3389/fmicb.2021.652520>
- Lopes, F.C., Castro, M.R., Barbosa, S.C., Primel, E.G., Martins, C.M.G. (2020). Effect of the UV filter, Benzophenone-3, on biomarkers of the yellow clam (*Amarilladesma mactroides*) under different pH conditions. *Mar Pollut Bull* 158. <https://doi.org/10.1016/j.marpolbul.2020.111401>
- Lusher, A., Hollman, P.C.H., Mendoza-Hill, Jeremy. (2017). Microplastics in fisheries and aquaculture : status of knowledge on their occurrence and implications for aquatic organisms and food safety (Technical Paper No./615). Rome: Food and Agriculture Organization of the United Nations (FAO). Available at: https://www.researchgate.net/publication/319644050_Microplastics_in_fisheries_and_aquaculture_Status_of_knowledge_on_their_occurrence_and_implications_for_aquatic_organisms_and_food_safety
- Millero, F. J., Woosley, R., Ditrolio, B., & Waters, J. (2009). Effect of Ocean Acidification on the Speciation of Metals in Seawater. *Oceanography*, 22(4), 72–85. <https://doi.org/10.5670/oceanog.2009.98>
- Nadella, S. R., Fitzpatrick, J. L., Franklin, N., Bucking, C., Smith, S., & Wood, C. M. (2009). Toxicity of dissolved Cu, Zn, Ni and Cd to developing embryos of the blue mussel (*Mytilus trossulus*) and the protective effect of dissolved organic carbon. *Comparative Biochemistry and*

- Physiology - Part C: Toxicology & Pharmacology*, 149, 340–348. <https://doi.org/10.1016/j.cbpc.2008.09.001>
- Orr, J. C., Fabry, V. J., Aumont, O., Bopp, L., Doney, S. C., Feely, R. A., Gnanadesikan, A., Gruber, N., Ishida, A., Joos, F., Key, R. M., Lindsay, K., Maier-Reimer, E., Matear, R., Monfray, P., Mouchet, A., Najjar, R. G., Plattner, G. K., Rodgers, K. B., ... Yool, A. (2005). Anthropogenic ocean acidification over the twenty-first century and its impact on calcifying organisms. *Nature*, 437, 681–686. <https://doi.org/10.1038/nature04095>
- Pierrot, D., Epitalon, J.-M., Orr, J.C., Lewis, E., and Wallace, D. W. R., MS Excel program developed for CO2 system calculations – version 3.0, (2021), GitHub repository. Access date: October 10th, 2024. https://github.com/dpierrot/co2sys_xl
- Pilson, M. E. Q. (2013). An introduction to the chemistry of the sea. *Second edition*. ed. Cambridge University Press. <https://doi.org/10.1017/CBO9781139047203>
- Plastics Europe (2024). Plastics – the fast facts 2024. Available at: https://plasticseurope.org/wp-content/uploads/2024/11/PE_TheFacts_24_digital-1pager.pdf. Access date: April 25th, 2025.
- Sekar, V., & Sundaram, B. (2026). Adsorption behaviour of accelerated UV aged PET and PP microplastics towards Pb (II) under varying pH, temperature, and salinity conditions. *Science of the Total Environment*, 1012, Article 181175. <https://doi.org/10.1016/j.scitotenv.2025.181175>
- Soroldoni, S., Martins, S. E., Castro, I. B., & Pinho, G. L. L. (2018). Potential ecotoxicity of metals leached from anti-fouling paint particles under different salinities. *Ecotoxicology and Environmental Safety*, 148, 447–452. <https://doi.org/10.1016/j.ecoenv.2017.10.060>
- Stockdale, A., Tipping, E., Lofts, S., & Mortimer, R. J. G. (2016). Effect of Ocean acidification on organic and inorganic speciation of trace metals. *Environmental Science & Technology*, 50, 1906–1913. <https://doi.org/10.1021/acs.est.5b05624>
- Strickland, J.D.H., Parsons, T.R. (1972). *A Practical handbook of seawater analysis*, Second edition. Ottawa: Fisheries Research Board of Canada. Available at: https://epic.awi.de/eprint/39262/1/Strickland-Parsons_1972.pdf
- Torres, F.G., Dioses-Salinas, D.C., Pizarro-Ortega, C.I., De-la-Torre, G.E. (2021). Sorption of chemical contaminants on degradable and non-degradable microplastics: Recent progress and research trends. *Science of the Total Environment* 757. <https://doi.org/10.1016/j.scitotenv.2020.143875>
- Turner, A., & Holmes, L. A. (2015). Adsorption of trace metals by microplastic pellets in fresh water. *Environmental Chemistry*, 12, 600–610. <https://doi.org/10.1071/EN14143>
- Underwood, A. J. (2011). *Experiments in ecology: Their logical design and interpretation using analysis of variance* (p. 524). Cambridge University Press.
- United States Environmental Protection Agency (USEPA) (2024). National Recommended Water Quality Criteria - Aquatic Life Criteria Table. Available at: https://www.epa.gov.translate.google.com/wqc/national-recommended-water-quality-criteria-aquatic-life-criteria-table?_x_tr_sl=en&_x_tr_tl=pt&_x_tr_hl=pt&_x_tr_pto=tc. Access date: May 8th, 2025.
- United States Environmental Protection Agency (USEPA) (2026). Aquatic Life Criteria - Copper. Available at: https://www.epa.gov.translate.google.com/wqc/aquatic-life-criteria-copper?_x_tr_sl=en&_x_tr_tl=pt&_x_tr_hl=pt&_x_tr_pto=tc. Access date: February 18th, 2025.
- Wang, F., Yang, W., Cheng, P., Zhang, Shuqi, Zhang, Shuwu, Jiao, W., & Sun, Y. (2019). Adsorption characteristics of cadmium onto microplastics from aqueous solutions. *Chemosphere*, 235, 1073–1080. <https://doi.org/10.1016/j.chemosphere.2019.06.196>
- Wang, Q., Zhang, Y., Wangjin, X., Wang, Y., Meng, G., & Chen, Y. (2020). The adsorption behavior of metals in aqueous solution by microplastics effected by UV radiation. *Journal of Environmental Sciences (China)*, 87, 272–280. <https://doi.org/10.1016/j.jes.2019.07.006>
- Wang, L., Wang, F., & Ma, Y. (2025). Combined effects of Ocean acidification and copper exposure on the polyps of moon jellyfish *Aurelia coerulea*. *Ecotoxicology*, 34, 712–724. <https://doi.org/10.1007/s10646-025-02869-w>
- Wang, X., Zhang, R., Li, Z., Yan, B. (2022). Adsorption properties and influencing factors of Cu (II) on polystyrene and polyethylene terephthalate microplastics in seawater. *Science of the Total Environment* 812. <https://doi.org/10.1016/j.scitotenv.2021.152573>
- Xiao, Z., Cao, L., Liu, J., Cui, W., Dou, S. (2023). pCO₂-driven seawater acidification affects aqueous-phase copper toxicity in juvenile flounder *Paralichthys olivaceus*: Metal accumulation, antioxidant defenses and detoxification in livers. *Science of the Total Environment* 858. <https://doi.org/10.1016/j.scitotenv.2022.160040>
- Zhang, J., Li, Z., Liang, J., Liu, G., Luo, Y., & Zhang, Q. (2025). Study on the mass concentration distributions of marine microplastics in estuaries and coastal areas. *Water*. <https://doi.org/10.3390/w17081136>
- Zheng, J., Li, Q., Zheng, X. (2023). Ocean acidification increases copper accumulation and exacerbates copper toxicity in *Amphioctopus fangsiao* (Mollusca: Cephalopoda): A potential threat to seafood safety. *Science of the Total Environment* 891. <https://doi.org/10.1016/j.scitotenv.2023.164473>
- Zou, J., Liu, X., Zhang, D., Yuan, X. (2020). Adsorption of three bivalent metals by four chemical distinct microplastics. *Chemosphere* 248. <https://doi.org/10.1016/j.chemosphere.2020.126064>
- Wang, K., Guo, C., Li, J., Wang, K., Liang, S., Wang, W., Wang, J. (2024). A critical review of the adsorption-desorption characteristics of antibiotics on microplastics and their combined toxic effects. *Environ. Technol. Innov.* <https://doi.org/10.1016/j.eti.2024.103729>
- Gao, Y., Gao, W., Liu, Y., Zou, D., Li, Y., Lin, Y., & Zhao, J. (2024). A comprehensive review of microplastic aging: Laboratory simulations, physicochemical properties, adsorption mechanisms, and environmental impacts. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2024.177427>
- Tang, S., Yang, X., Zhang, T., Qin, Y., Cao, C., Shi, H., & Zhao, Y. (2022). Adsorption mechanisms of metal ions (Pb, Cd, Cu) onto polyamide 6 microplastics: New insight into environmental risks in comparison with natural media in different water matrices. *Gondwana Research*, 110, 214–225. <https://doi.org/10.1016/j.gr.2022.06.017>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.